

Class 10 Science - Chapter 8

Heredity (Hinglish Notes)

PART A - CORE BASICS (MISS MAT KARNA)

Ch7 + Class 9 ki yeh cheezein pehle pakki karo, warna Ch8 samajh nahi aayega.

1. Cell, Nucleus, DNA, Chromosome, Gene

Cell ke nucleus me CHROMOSOME hote (dhaage jaise).

Chromosome DNA + protein ka bana. DNA me jeev ki saari instructions.

GENE = DNA ka chhota tukda jo ek trait (lakshan) ko control karta.

Jaise: height ka gene, aankh ke rang ka gene.

Human me 23 jodi (=46) chromosome hote (har parent se 23-23).

2. Trait, Gamete, Fertilisation (Ch7 link)

Trait (lakshan) = jeev ki koi pehchaan (height, rang, beej ka aakar).

Gamete = reproductive cell (sperm/egg) - isme aadhe (23) chromosome.

Fertilisation: sperm(23) + egg(23) -> zygote(46). Isi se bachche ko dono parents ke gene milte hain.

3. Variation aur Heredity ka idea

Heredity (anuvanshikta) = parents se bachchon me traits ka transfer.

Variation = ek species ke jeevon me antar.

Bachcha parents jaisa hota PAR bilkul same nahi (variation ki wajah se).

4. Dominant/Recessive ka simple idea

Har trait ke liye 2 gene copy (ek ma se, ek pita se).

Agar 2 alag copy ho to jo DIKHTA hai = DOMINANT (haavi),
jo chhup jaata = RECESSIVE (dabba hua).

5. Probability (Maths link)

Probability = kisi cheez ke hone ki sambhavna (0 se 1, ya % me).

Sikka uchaalo -> head ya tail, dono ka chance 1/2 (50%).

Genetics me ratio nikalne ke liye yahi probability kaam aati.

-- END OF CORE BASICS (RED) PAGE --

PART B - CHAPTER KA MAIN CONTENT

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[GREEN = board exam me baar-baar poocha jaata hai]

1. Heredity aur Mendel

Heredity = ek generation se agli me traits (lakshan) ka jaana.

Gregor MENDEL ("Father of Genetics") ne MATAR (pea plant, *Pisum sativum*) par experiment kiye -> heredity ke niyam diye.

Matar kyun chuna? - jaldi badhta, alag-alag clear traits (lamba/bauna, peela/hara beej), self+cross pollination dono possible.

2. Key Terms [VERY IMPORTANT - RATTA]

Gene : trait control karne wala DNA ka tukda.

Allele : ek gene ke alag roop (jaise T = lamba, t = bauna).

Dominant : jo trait dikhta hai (capital letter, T).

Recessive : jo dab jaata, sirf akela hone par dikhta (small letter, t).

Genotype : gene ka combination (TT, Tt, tt).

Phenotype : jo bahar dikhta (lamba ya bauna).

Homozygous : dono allele same (TT ya tt) - "pure".

Heterozygous: dono allele alag (Tt) - "hybrid".

F1 generation: pehli santaan (parents = P). F2 = F1 ka aapas me cross.

3. Monohybrid Cross (ek trait) [TOP EXAM AREA]

Mendel ne pure lamba (TT) x pure bauna (tt) cross kiya:

P: TT x tt

F1: sab Tt -> SAARE LAMBE (dominant T dikha; t chhup gaya).

Phir F1 (Tt) x F1 (Tt) cross kiya:

Punnett square:

	T	t
T	TT	Tt
t	Tt	tt

F2 genotype: 1 TT : 2 Tt : 1 tt.

F2 phenotype: 3 LAMBE : 1 BAUNA (RATIO 3:1).

Conclusion: recessive trait (bauna) F1 me gayab hokar F2 me wapas aaya -> gene jodon (pairs) me hote, alag-alag santaan me baant-te.

4. Dihybrid Cross (do trait ek saath) [VERY IMPORTANT]

Mendel ne 2 traits ek saath dekhe: beej ka rang (Yellow Y / green y) aur aakar (Round R / wrinkled r).

P: RRYy (round-yellow) x rryy (wrinkled-green)

F1: sab RrYy -> round-yellow (dono dominant dikhe).

F2 phenotype ratio = 9 : 3 : 3 : 1

(9 round-yellow : 3 round-green : 3 wrinkled-yellow : 1 wrinkled-green).

Important: F2 me NAYE combinations (round-green, wrinkled-yellow) bhi aaye -> matlab do traits SWATANTRA roop se (independently) inherit hote hain.

5. Mendel ke Niyam (Laws) [IMPORTANT]

- (1) Law of Dominance: do alag allele me se ek (dominant) hi dikhta hai.
- (2) Law of Segregation: gamete bante waqt allele ka jodi alag (segregate) ho jaata - har gamete me ek hi allele jaata.
- (3) Law of Independent Assortment: alag-alag traits ke gene ek doosre se swatantra roop se inherit hote (dihybrid cross se pata chala).

6. Sex Determination in Humans [TOP EXAM AREA]

Human me 23 jodi chromosome - 22 jodi AUTOSOMES + 1 jodi SEX CHROMOSOME.

Female: XX. Male: XY.

Mother hamesha X deti (egg me sirf X).

Father X ya Y de sakta (sperm me X ya Y).

Agar father ka X mila -> XX -> LADKI.

Agar father ka Y mila -> XY -> LADKA.

IMPORTANT CONCLUSION: bachche ka ling (gender) FATHER (ke sperm) par depend karta, mother par NAHI. (Yeh samaj ka galat-fehmi todta - exam fav.)

7. Acquired vs Inherited traits

Inherited trait: gene se mila, agli generation me jaata (aankh ka rang).

Acquired trait: jeevan me khud se aaya (jaise weightlifting se muscle, kata hua haath) - yeh DNA/gene me nahi jaata, isliye agli generation me NAHI jaata. (Eg: chuhe ki poonch kaat-te raho, bachche ki poonch normal hi rahegi.)

8. Quick Revision - One Liners

- Mendel = father of genetics; matar par experiment.
- Monohybrid F2 = 3:1; Dihybrid F2 = 9:3:3:1.
- Genotype (TT/Tt/tt) gene; Phenotype (dikhne wala).
- Female XX, Male XY; bachche ka gender FATHER decide karta.
- Acquired trait agli generation me nahi jaata.

PART C - SOLVED EXAMPLES (HARDEST -> EASIEST)

Example 1 (HARDEST) - Monohybrid cross poora + ratio

Q: Pure lamba (TT) ko pure bauna (tt) se cross karo. F1 aur F2 ka genotype + phenotype ratio nikaalo. Recessive trait F2 me wapas kyun aaya?

Solution:

- P: TT x tt -> F1: sab Tt (saare LAMBE; T dominant).
- F1 x F1 (Tt x Tt) Punnett:
T t / T t -> TT, Tt, Tt, tt.
- F2 genotype = 1 TT : 2 Tt : 1 tt.
- F2 phenotype = 3 lamba : 1 bauna (3:1).
- Bauna (tt) F1 me chhup gaya tha (t recessive), F2 me jab dono t mile (tt) to wapas dikha -> gene jodon me hote aur alag-alag baant-te.

Example 2 - Dihybrid F2 ratio + nayi combination

Q: RRYy x rryy cross me F1 aur F2 phenotype ratio? Naye combinations kya batate?

Solution:

- F1: sab RrYy (round-yellow).
- F2 phenotype = 9 round-yellow : 3 round-green : 3 wrinkled-yellow : 1 wrinkled-green (9:3:3:1).
- Naye combinations (round-green, wrinkled-yellow) batate ki do traits SWATANTRA roop se inherit hote (Law of Independent Assortment).

Example 3 - Sex determination

Q: Bachche ka ling (gender) kaun determine karta - ma ya papa? Samjhao.

Solution:

- Mother sirf X deti (XX). Father X ya Y deta (XY).
- Father ka X -> XX (ladki); father ka Y -> XY (ladka).
- Isliye bachche ka gender FATHER ke sperm par depend karta, mother par nahi.

Example 4 - Acquired vs inherited

Q: Agar koi insaan jeevan bhar gym karke muscle banaye, kya woh muscle uske bachche me aayega? Kyun?

Solution:

- Nahi. Yeh ACQUIRED trait hai (jeevan me khud banaya), gene/DNA me change nahi hua. Sirf inherited (gene wale) traits hi agli generation me jaate.

Example 5 (EASIEST) - Genotype vs Phenotype

Q: Genotype aur phenotype me antar batao.

Solution:

- Genotype = gene combination (TT, Tt, tt).
- Phenotype = jo bahar dikhta (lamba ya bauna).

PART D - SELF TEST (EASY -> HARD)

EASY (1 mark)

- Q1. Heredity aur gene ka matlab.
- Q2. Dominant aur recessive trait me antar.
- Q3. Human me kitne chromosome? Sex chromosome female/male me kya?
- Q4. Mendel ne kaun se plant par kaam kiya aur kyun?

MEDIUM (2-3 marks)

- Q5. Genotype, phenotype, homozygous, heterozygous samjhao.
- Q6. Monohybrid cross F2 ratio (Punnett square sahit).
- Q7. Acquired aur inherited trait me antar + example.
- Q8. Mendel ke 3 niyam (laws) likho.

HARD (3-5 marks, board favourite)

- Q9. Monohybrid cross: P se F2 tak + 3:1 ratio + recessive wapas kyun.
- Q10. Dihybrid cross F2 = 9:3:3:1 + independent assortment samjhao.
- Q11. Sex determination in humans poora (XX/XY, father decide).
- Q12. Mendel ne kaise prove kiya ki traits gene (pairs) ke roop me inherit hote.

ANSWER HINTS (PART D)

- A1. Heredity = parents se traits ka transfer; Gene = trait wala DNA tukda.
- A2. Dominant dikhta (T); recessive dab jaata, akela hone par dikhta (t).
- A3. 23 jodi (46); female XX, male XY.
- A4. Matar (pea); jaldi badhta, clear traits, self+cross pollination.
- A5. Genotype (gene), phenotype (dikhne wala), homozygous (TT/tt), hetero (Tt).
- A6. $Tt \times Tt \rightarrow 1TT:2Tt:1tt \rightarrow$ phenotype 3:1.
- A7. Inherited gene se (agli gen me jaata); acquired khud banaya (nahi jaata).
- A8. Dominance, Segregation, Independent assortment.
- A9. $TT \times tt \rightarrow F1 Tt$ (lamba) $\rightarrow F2$ 3 lamba:1 bauna; recessive tt me wapas.
- A10. $RRYY \times rryy \rightarrow F1 RrYy \rightarrow F2$ 9:3:3:1; naye combinations = independent.
- A11. Mother X, father X/Y; X->ladki, Y->ladka; father decide karta.
- A12. Recessive trait F1 me gab F2 me wapas $\rightarrow$ gene pairs me, segregate hote.

"Padhai tabhi pakki jab khud likh ke test do." - All the best, Krishna!

Generated by Kiro for Krishna1k
